Task 7 – Recurrence solution by iteration

$T(n) = T(n/2) + c$

$= T(n/4) + 2c$

$= T(n/8) + 3c$

$= T(n/2^j) + jc$

Stop when $n/2^j <= 1$, so $j >= \log_2(n)$. Therefore $T(n) = T(1) + c \log_2(n) = \Theta(\log n)$.

The iteration method applies because the problem size is halved at each recursive call and the work outside the call is constant. The result explains why recursive binary search scales well for large sorted datasets.